RL for Real
Hands-on assignments

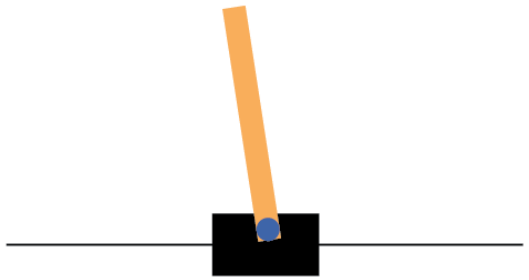
Ido Greenberg

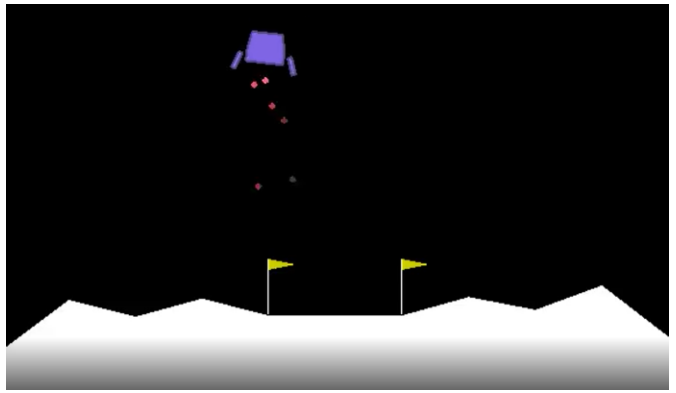
2026

Policy Gradient warmup: From REINFORCE to PPO

Goals

- Review PG's flow end-to-end (**important!**)
- Implement core PG functions (~25 lines in total)
- Train your own agent on classic control problems
- Visualize & compare REINFORCE vs. PPO





Exercise repo

- https://github.com/ido90/toy-ppo
- Follow `Setup` instructions carefully
- Follow `Exercise` instructions

Setup

Tuesday-Thursday

Setup

- Desks at Floors 3-4
- Make sure ethernet cables are **disconnected**
- Work remotely on either of the two:
  - Your remote machine
  - Cloud compute via BREV [onboarding instructions]

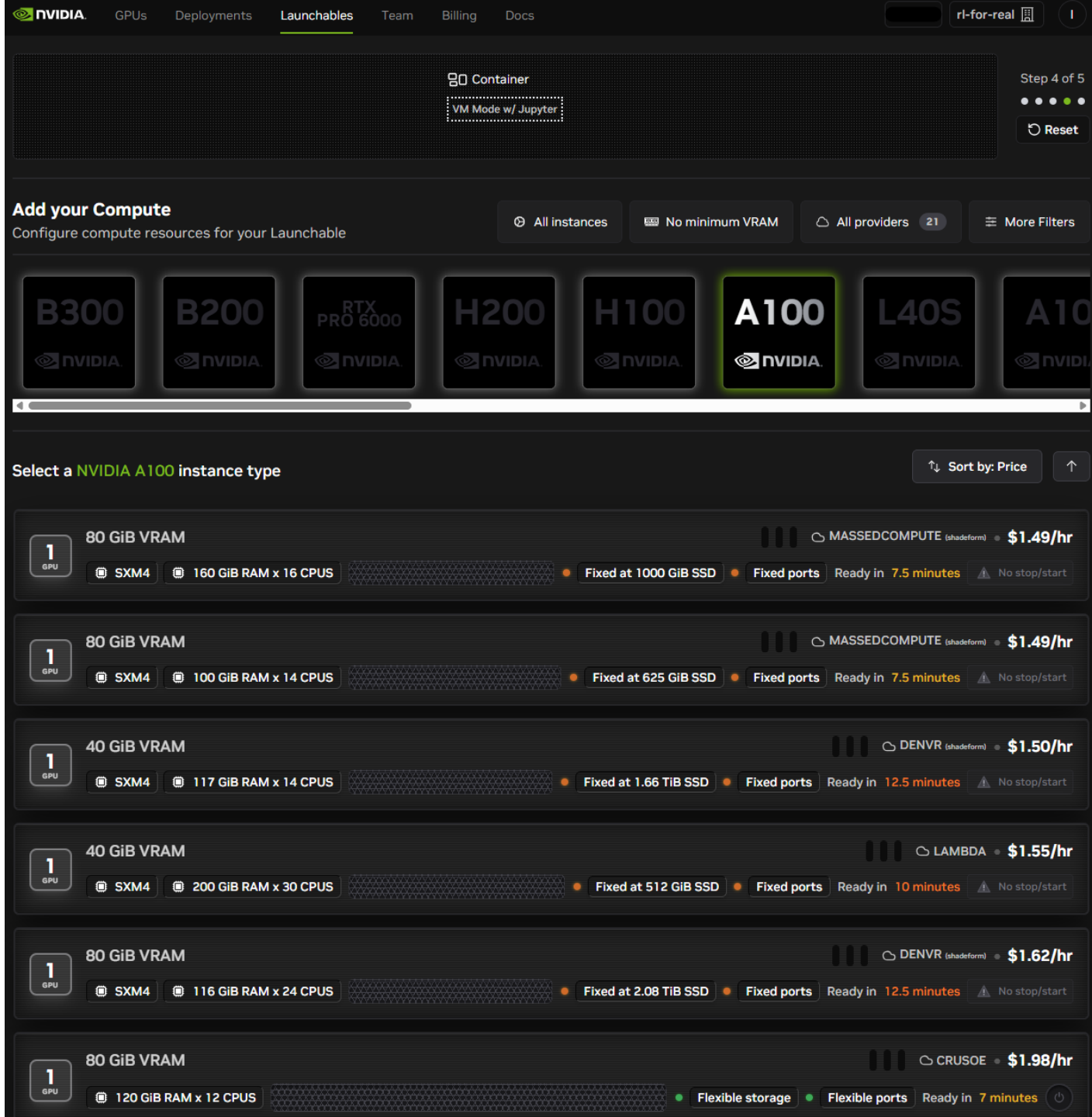
BREV

- NVIDIA's platform to connect compute providers and users
- **Launchable / Deployment** = a machine's setup to be deployed
- **Instance** = a specific machine

- Recommended editor: vscode / cursor
- **Delete your instance once finished (you pay till you delete!)**

- BREV: https://brev.nvidia.com/
- Instructions for the course:

https://docs.google.com/document/d/1JgvN5PXLb2kavnYArNliwEzkA-plDtKgEbskvNhndrk/



HW requirements & budget

Day	Exercise	Compute time	Recommended HW	Estimated BREV cost
Sunday (11.10)	Toy Policy Gradient	1.5h	CPU is enough	0$
Tuesday (13.10)	LLM Sentiment	3h	A6000	2$
Wednesday (14.10)	LLM Reasoning	7.5h	A100	15$
Thursday (15.10)	Robotics	6h	RTX Pro 6000 / A100	9$
Total				**26$**

- Each student receives a BREV coupon of **60$**
- If your own machine's HW is sufficient, you can choose to use it

LLM Sentiment

Goals

- Fine-tune your LLM via RL to be more positive
  - Or negative ☺ (`--negate_reward`)
- Experience challenges with reward design and resolve them
- Try to force your own will upon the model
  - Compare RL vs. prompt-engineering

Recommended HW

- **Small scale: GPT-2, 10-minute train cycles**
- GPU with 8GB VRAM is enough
- On BREV: A6000 is good value for money

Exercise repo

- https://github.com/ido90/rl-sentiment-exercise
  - Using TRL (Transformer Reinforcement Learning)
- Follow `Setup` instructions carefully
- Follow `Exercises` instructions

LLM Reasoning

Overview

- Fine-tune QWEN (2.5-3B-Instruct) via RL to solve reasoning tasks
- Benchmark: Reasoning-Gym
- Training infra: TRL
- Exercise repo: https://github.com/ido90/rl-for-real

- **Goal: get the best test score you can**

HW

- Recommended: A100 with 80GB VRAM
- Minimal: 40GB VRAM

Reasoning Gym

- Benchmark for RL for LLM-reasoning

- *Domain* (e.g. algorithms / games)
  - *Task* (e.g. maze / rush hour)
    - *Instance* – automatically generated (e.g. maze shape / marker types)
    - Task configuration defines distribution of instances (and difficulty)

- Required format: “<think> … </think> <answer> … </answer>”

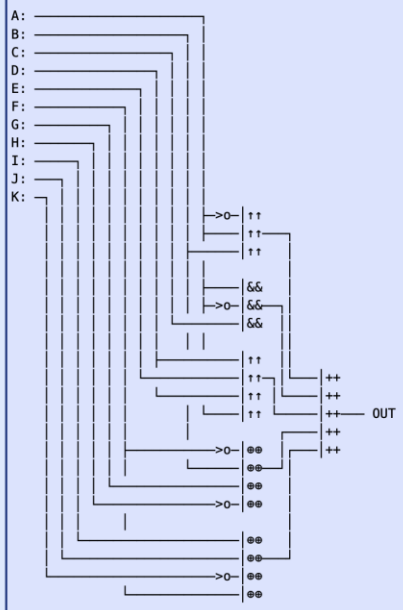
Reasoning Gym

- Examples:

Logic

Circuit logic

Question: Below is a logic circuit.



Legend for gates:
&&: AND | ↑↑: NAND | ⊕⊕: XOR | >o: Negate | ++: OR

Given the following input assignments:
A = 0; B = 0; C = 0; D = 1; E = 1; F = 0; G = 1;
H = 1; I = 0; J = 0; K = 0

What is the final output?

Answer: 1

Cognition

Binary matrix

Given a square matrix, your job is to find the Manhattan distance of the nearest 0 for each cell.

The output should be a matrix of the same size as the input matrix, where each cell contains the distance to the nearest 0.

```
0 1 1 0 1 0 1 0 1
0 0 1 1 1 0 1 1 1
1 1 1 1 0 1 0 1 0
0 1 0 1 1 1 1 1 0
1 0 1 0 1 1 1 1 1
1 0 1 1 1 0 0 1 0
1 1 1 1 1 1 1 0 1
1 1 0 0 0 0 1 1 0
0 1 0 1 1 0 1 1 1
```

Answer:

```
0 1 1 0 1 0 1 0 1
0 0 1 1 1 0 1 1 1
1 1 1 1 0 1 0 1 0
0 1 0 1 1 2 1 1 0
1 0 1 0 1 1 1 2 1
1 0 1 1 1 0 0 1 0
2 1 1 1 1 1 1 0 1
1 1 0 0 0 0 1 1 0
0 1 0 1 1 0 1 2 1
```

Figlet fonts

Question: What word does this say?

```
8888ba.88ba      88888888b    dP          dP    .d88888b     .d88888b      .d888888
88  `8b  `8b    88           88          88    88.    "'    88.    "'    d8'    88
88   88   88    a88aaaa      88          88    `Y88888b.    `Y88888b.    88aaaaa88a
88   88   88     88          88          88          `8b          `8b    88     88
88   88   88     88          88          88    d8'   .8P    d8'   .8P    88     88
dP   dP   dP     88888888P   88888888P   dP     Y88888P      Y88888P     88     88
```

Answer: MELISSA

Games

Rush hour

Question:
Move the red car (AA) to the exit on the right.
Specify moves in the format: 'F+1 K+1 M-1 C+3 H+2 ...',
where the letter is the vehicle and +/- number is spaces
to move right/left or down/up. Walls are marked with an 'x'.

Cars cannot move through walls, and walls cannot be moved.
A car oriented vertically can only move up and down,
a car oriented horizontally can only move left and right.

Board:

```
BB.IKx
CCCIK.
GAAJ..
G.HJDD
..HEEL
.FF.xL
```

Answer:
F-1 G+1 A-1 H-1 E-2 J+2 D-1 L-3 D+1 J-2 E+3 H+2 A+1 J+2 ...

Rubik's cube

Question: You see a size 3 Rubik's cube. It is arranged this:

```
          B  Y  R
          B  Y  W
          B  Y  Y
 R  R  R  Y  G  G  O  O  G  Y  B  W
 R  R  R  Y  G  G  O  O  W  G  B  W
 R  R  Y  B  B  B  W  O  W  G  B  W
          O  O  O
          G  W  Y
          G  W  O
```

Please provide a solution to solve this cube using Singmaster notation. Do not combine any steps, for instance, do not write 'U2', and instead write 'U U'.

Answer:
L' D D B' D D R R U' F' U F U U B U' B' R' U U R U' F U U F'

Reasoning Gym

- More realistic examples for QWEN-2.5-**3B**:

```
Question: Navigate from '3' (start) to 'z' (goal):

```
>>>>>>>>>
>eeee>e>>
>ee>>>>>>
>eeeeee>>
>e>ee>>e>
>>ez>3e>>
>eee>e>e>
>eeeee>e>
>>>>>>>>>
```
Legend: '>' = Wall, 'e' = Passage

What is the minimum number of steps to reach the goal?
Give only the number of steps as your final answer, no oth
Answer: 6
```

```
Question: John is married to Isabella. They have a child called Edward. Edward is married t

What is Isabella to Edward? Respond only with the word that describes their relationship.
Answer: mother
```

```
Question: Remove all numbers larger than 18.236 in this list: ['-42.8', '91.88', '34']
Return the new list in the same format.
Answer: ['-42.8']
```

Provided train settings

Setting	Config file	Train time	Train tasks	Test tasks	Results
Toy	syllogism_toy.yaml	<10m	syllogism	Same	13% → 73%
Full	multi_tasks.yaml	~40m	number_filtering, string_insertion, family_relationships, maze	Same + base_conversion	11% → 28%

Goal: improve test accuracy on the Full setting

Task examples:
https://github.com/open-thought/reasoning-gym/blob/main/GALLERY.md

How to improve the score?

- **Any way you like**
  - Reward shaping
  - Hyper-parameter tuning (TRL already includes MANY features)
  - More suggestions and details at the repo's README

- Schedule note:
  - **12:00-14:00:** you eat lunch – models keep training
  - Keep training if disconnected: **`tmux new -s train`**

Evaluation

- **Challenge I: best overall test score**
  - Subject to eligible conditions
  - Metric: highest `eval/accuracy` during training

- **Challenge II: coolest training results**
  - Freestyle – no eligibility conditions
    - Different tasks, larger models, creative constraints
  - Metric: voting based on 2min presentation

Eligible conditions

- In `multi_tasks.yaml`, Do not change:
  - `model_name_or_path`
  - `eval_datasets`
  - `datasets`' keys (you may change train task parameters, but don't use other training tasks)
- Do change:
  - `eval_num_samples` = 200
- Do not hack eligible conditions

Evaluation

- Working in groups is allowed
- Sharing insights is ~~allowed~~ recommended

- Submission:
  - https://docs.google.com/forms/d/e/1FAIpQLSeibLY88sVuez9cVxqzOdpOA6GYlsS3v9fhHkv1R7PpdoJm3w/viewform
  - Deadline: **14:50**

Wrapping up

- By **14:50**, make sure to:
  - Save any code/data/models you like
  - Delete all BREV instances
  - Write a few slides (if relevant)

- Presentations
  - Overall score winner (4min)
  - Freestyle contenders (2min)
  - Others are welcome to share insights if time permits